

# claude-windows / 6a82308f-69d6-43a8-9ea4-d8eaf497253d

## Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-index\6a82308f-69d6-43a8-9ea4-d8eaf497253d\subagents\workflows\wf_3d536535-241\agent-a1de4494be1c6b209.jsonl`  
- SHA-256: `7c8bc433929c763866d7c04fee6361ad2c2861193d0056829c085352ffaed319`  
- Source modified: `2026-06-16T03:24:58+00:00`  
- Imported at: `2026-07-05T16:48:27+00:00`  
- project: `wf_3d536535-241`  
- session_id: `6a82308f-69d6-43a8-9ea4-d8eaf497253d`
```

## Transcript

1. user (2026-06-16T03:22:59.899Z)

### Adversarial Claim Verifier (voter 2/3)

Be SKEPTICAL. Try to REFUTE this claim. 2/3 refutations kill it.

### Research question

Research question: **What concrete, recent (2022-2025) techniques and evaluation practices should a local, commercially-licensed badminton rally-detector adopt to close the quality gap with a strong cloud VLM (Gemini), given a tiny labeled set (~6 - ~16 golden videos)?** Produce an adversarially-verified, cited report organized around the four dimensions below. Prefer specific named methods, papers (arXiv/venue), and reported numbers over generic advice; flag anything you cannot verify.

CONTEXT (so you target the real gap, not generic ML):

- System: a local pipeline detects the shuttle per-frame (WASB / TrackNet, MIT-licensed) ? a heuristic turns the trajectory into candidate rally windows ? today an optional Gemini "handoff" confirms rallies. We just measured a fully-local (no-AI) run vs Gemini on 3 matches: local emits far FEWER but MUCH LONGER windows (mean 65s vs 6.1s; one window spanned a whole 174s clip) ? it does NOT miss activity, it FAILS TO SEGMENT. Root cause (code-confirmed): a frame is "active" iff the shuttle is visible AND moving above a small per-second-normalized velocity threshold; active frames within a 2 s gap are merged; there is NO max-window cap and NO inactivity/boundary split. So "shuttle is moving" is conflated with "rally in progress," and dead-time + serves + knock-ups bridge rallies into mega-windows. Shuttle tracking itself is excellent (96-99% per-frame visibility).

- HARD CONSTRAINTS: weights/data/code must be MIT/Apache-2.0/BSD only (no viral/NC/custom licenses, reject Gemma-3/Llama-4-MAU-cap); we may NEVER train owned weights on paid-LLM (Gemini/OpenAI/Anthropic) outputs (terms/copyright), though paid LLMs may be used at inference; minors excluded from any training pool; runs on a single local GPU.

- WHAT WE ALREADY HAVE (do NOT re-survey the basics ? go deeper/newer): the architecture skeleton is known to be standard Temporal Action Localization/Segmentation (proposal+scorer) + late/score-level fusion + stacked generalization; in-domain prior art (MonoTrack, ShuttleSet, See et al. 2024/25 non-broadcast badminton rally start/end, Ghosh et al. point-segmentation, tennis/TT play-break state machines) is already cited. We have a working eval harness: leave-one-video-out (grouped by match), temporal-IoU F1 @0.5, F1@{0.1,0.25,0.5}, mAP@tIoU, segment-count-ratio, bootstrap 95% CIs, paired exact sign-test, Platt/isotonic calibration + Brier/ECE, a tiny numpy LogisticRegression scorer over per-window feature columns, an experiment leaderboard, and an "eval+serve" guard (a cue measured +0.074 F1 on permissive proposal windowing but ?0.008 on the coarser served windowing ? reverted). Available but default-OFF per-window signals: net-crossing count (rally gate), 5 person/court features, 3 audio-hit features.

THE FOUR DIMENSIONS:

1) WINDOWING & BOUNDARIES (Temporal Action Localization/Segmentation): beyond proposal+score ? what are the best **recent** methods for (a) turning a dense per-frame signal into well-segmented

actions, (b) explicitly controlling OVER-segmentation/merging, (c) boundary detection/regression and splitting merged segments? Cover e.g. MS-TCN/MS-TCN++, ASRF (action-boundary regression), BMN/BSN boundary-matching, ActionFormer/TriDet/TemporalMaxer (actionness + boundary heads), smoothing/boundary losses, and the standard metrics for over-segmentation (segmental F1@k, edit/Levenshtein score, mAP@tIoU). Which transfer to a 1-D shuttle-trajectory + auxiliary-cue setting on tiny data?

2) SHUTTLE-TRAJECTORY ? RALLY SIGNAL PROCESSING (racket-sports specific): how do published systems convert ball/shuttle tracking into rally/point segments and distinguish "in-play" from dead-time? Cover rally-state machines, serve/let detection, net-crossing & two-sided-occupancy cues, hit/contact (audio + kinematic) detection, and fps-invariant trajectory features. What features best separate "rally" from "shuttle merely moving"?

3) SMALL-DATA STRATEGIES: with ~6?16 labeled videos, what gives the most quality per label ? active learning (which videos/segments to label next; uncertainty/diversity/core-set), semi-/weakly-/point-supervised TAL (e.g. SF-Net, timestamp supervision), self-training/pseudo-labeling, and data-efficient TAL? Separately: the legally-clean ways to exploit a strong teacher (Gemini) at INFERENCE for weak/pseudo-labels or as an evaluation oracle WITHOUT its outputs becoming training data for owned weights ? what does the literature say about teacher-student / distillation boundaries and label-noise handling, and what governance distinctions matter?

4) EVALUATION METHODOLOGY & RIGOR at small n: trustworthy measurement with ~6?16 videos ? bootstrap/permutation tests, paired sign-tests, leave-one-group-out vs RL00CV bias, calibration (reliability, ECE), the over-segmentation-blindness of plain tIoU, shadow/A-B evaluation, regression gates, and concrete pitfalls (label noise, annotator agreement/IAA, multiple-comparison/over-fitting traps, eval-vs-serve drift). What metrics + protocols should a small-data sports-segmentation leaderboard standardize on?

Deliverable: a structured, cited report with, per dimension, the 3?5 highest-leverage techniques that APPLY to our constrained setting (small data, local, license-clean, 1-D trajectory + cues + learned scorer), each with: what it is, why it fits our over-merge/small-data problem, the expected difficulty/payoff, and the citation. Call out which ideas are low-effort high-confidence wins vs research bets. Be explicit about anything unverified or license-incompatible.

## Claim under review

"ASRF reduces over-segmentation by adding a complementary network branch that directly regresses action-boundary locations, and this boundary-refinement approach is model-independent ? it can be applied to the output of any temporal segmentation model (i.e. usable as a post-hoc splitter on an existing dense signal)."

**\*\*Source:\*\*** <https://arxiv.org/pdf/2210.10352> (primary)

**\*\*Supporting quote:\*\*** "ASRF [97] refines boundaries with a complementary network branch that directly regresses boundary locations. This approach is model-independent and applicable to any temporal segmentation output."

## Checklist

1. Is the claim actually supported by the quote, or is it an overreach/misread?
2. WebSearch for contradicting evidence ? does any credible source dispute or heavily qualify this?
3. Is the source quality sufficient for the claim's strength? (extraordinary claims need primary sources)
4. Is the claim outdated? (check dates ? old claims about fast-moving fields are suspect)
5. Is this a marketing claim / press release / cherry-picked benchmark / forum speculation?

**\*\*refuted=true\*\*** if: unsupported by quote / contradicted / low-quality source for strong claim / outdated / marketing fluff.

**\*\*refuted=false\*\*** ONLY if: claim is well-supported, current, and source quality matches claim strength.

Default to refuted=true if uncertain.

Structured output only. Evidence MUST be specific.

2. assistant (2026-06-16T03:23:02.413Z)

I'll verify this claim about ASRF. Let me start by examining the cited source and checking the actual ASRF paper.

### 3. assistant (2026-06-16T03:23:04.952Z)

Now let me fetch the cited source PDF to verify the supporting quote actually exists there.

### 4. user (2026-06-16T03:23:10.971Z)

I cannot reliably extract the title, topic, or specific content about ASRF from the provided material. The content appears to be a PDF file with heavily encoded/compressed data (FlateDecode and DCTDecode streams), making it impossible to parse the readable text accurately.

To properly answer your questions about:

- The paper's title and topic
- Whether ASRF is mentioned
- Specific statements about boundary regression and model-independence

I would need the document in a text-readable format (plain text, HTML, or properly decompressed PDF content) rather than the binary/compressed stream data provided.

Could you supply the paper in a more accessible format?

[Binary content (application/pdf, 1.8MB) also saved to C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\tool-results\webfetch-1781579999760-41alwg.pdf]

### 5. assistant (2026-06-16T03:23:13.275Z)

The PDF didn't parse via WebFetch. Let me check the arXiv abstract page to identify what 2210.10352 actually is, and look at the saved PDF directly.

### 6. user (2026-06-16T03:23:15.505Z)

pdftoppm failed: Command 'pdftoppm' not found or is in an unsafe location (current directory)

### 7. user (2026-06-16T03:23:18.781Z)

## Paper Summary

**Title:** "Temporal Action Segmentation: An Analysis of Modern Techniques"

**Authors:** Guodong Ding, Fadime Sener, Angela Yao

**Abstract:** The paper addresses temporal action segmentation in videos, aiming to "densely identify video frames in minutes-long videos with multiple action classes." It systematizes recent TAS methods, examining task definitions, benchmarks, supervision types, and evaluation metrics. The authors focus on two core techniques: frame representation and temporal modeling across different supervision levels.

**Topic:** This is a survey paper on temporal action segmentation (TAS), a computer vision task for understanding and classifying actions in long videos frame-by-frame.

**ASRF and Boundary Regression:** Based on the content provided, the abstract does not mention ASRF (Action Segment Refinement Framework) or boundary regression. The page excerpt focuses on the survey's scope but doesn't detail specific methods or techniques like these. To determine if the full paper discusses these approaches, one would need to access the complete document.

### 8. assistant (2026-06-16T03:23:22.057Z)

So the cited source `2210.10352` is the survey **"Temporal Action Segmentation: An Analysis of Modern Techniques"** by Ding, Sener, Yao (a TPAMI/arXiv survey). This is a legitimate, high-quality survey paper ? good source quality. The claim attributes a specific characterization of

ASRF to it. Let me extract the actual PDF text to find what this survey actually says about ASRF.

## 9. user (2026-06-16T03:23:27.721Z)

--- ASRF context ---

9 60.2 77.5 74.8 [91] UVASt 2022 I3D 92.7 91.3 81.0 92.1 80.2 76.9 71.5 58.0 77.1 69.7 Refine  
[110] MTDA + MS-TCN 2020 I3D 90.5 88.4 76.2 85.8 80.0 74.2 68.6 56.5 73.6 71.0 [111] SSTDA + MS-  
TCN 2020 I3D 90.0 89.1 78.0 86.2 79.8 75.0 69.1 55.2 73.7 70.2 [79] GTRM + MS-TCN\*\* 2020 I3D - -  
- - - 57.5 54.0 43.3 58.7 65.0 [96] BCN + MS-TCN 2020 I3D 88.5 87.1 77.3 84.4 79.8 68.7 65.5  
55.0 66.2 70.4 [97] ASRF + MS-TCN 2020 I3D 89.4 87.8 79.8 83.7 77.3 74.3 68.9 56.1 72.4 67.6  
[112] G2L + MS-TCN 2021 I3D 89.9 87.3 75.8 84.6 78.5 74.9 69.0 55.2 73.3 70.7 [95] FIFA + MS-TCN  
2021 I3D - - - - 75.5 70.2 54.8 78.5 68.6 [95] FIFA + UVASt [91] 2022 I3D 82.9 79.4 64.7 90.5  
69.8 76.9 71.5 58.0 77.1 69.7 [113] DTL + MS-TCN 2022 I3D - - - - 73.0 67.7 54.4 71.6 72.3  
[114] DPRN + MS-TCN 2022 I3D 92.9 92.0 82

--- ASRF context ---

res. Additionally, DiffAct proposes a masking strategy that jointly uses position, boundary, and relation priors of human actions to enhance the segmentation results. Several post-processing approaches target improving the action boundaries. BCN [96] introduces a module for MS-TCN that features a pooling operator to smooth noisy low-confidence boundary predictions with clean, confident ones. ASRF [97] refines boundaries with a complementary network branch that directly regresses boundary locations. This approach is model-independent and applicable to any temporal segmentation output. DPRN [114] proposes a divide-and-conquer approach to maximize frame-wise classification accuracy and reduce over-segmentation errors. UARL [115] estimates the uncertainty stemming from ambiguous boundaries.

## 10. assistant (2026-06-16T03:23:34.766Z)

This is a critical finding. The exact supporting quote **does appear verbatim** in the survey (2210.10352): **"ASRF [97] refines boundaries with a complementary network branch that directly regresses boundary locations. This approach is model-independent and applicable to any temporal segmentation output."**

So the quote is genuine and accurately transcribed. Now the harder adversarial test: **is the survey's characterization itself correct?** The claim's strong part is that ASRF is a **\*post-hoc splitter applicable to ANY temporal segmentation output\***. Let me check the primary ASRF paper (Ishikawa et al., WACV 2021, "Alleviating Over-segmentation Errors by Detecting Action Boundaries", ref [97]) to see whether ASRF is actually model-independent/post-hoc, or whether it's a jointly-trained dual-branch architecture (which would make "post-hoc splitter on an existing dense signal" an overreach).

## 11. user (2026-06-16T03:23:42.958Z)

Web search results for query: "ASRF Alleviating Over-segmentation Errors Detecting Action Boundaries Ishikawa boundary regression branch jointly trained"

Links: [{"title": "Alleviating Over-segmentation Errors by Detecting Action Boundaries | IEEE Conference Publication | IEEE Xplore", "url": "https://ieeexplore.ieee.org/document/9423184/"}, {"title": "Alleviating Over-segmentation Errors by Detecting Action Boundaries | DeepAI", "url": "https://deepai.org/publication/alleviating-over-segmentation-errors-by-detecting-action-boundaries"}, {"title": "Alleviating Over-segmentation Errors by Detecting Action Boundaries", "url": "https://yiskw713.github.io/asrf/"}, {"title": "Alleviating Over-segmentation Errors by Detecting Action Boundaries", "url": "https://arxiv.org/pdf/2007.06866"}, {"title": "asrf/evaluate.py at main · yiskw713/asrf", "url": "https://github.com/yiskw713/asrf/blob/main/evaluate.py"}, {"title": "GitHub - yiskw713/asrf: Code for 'Alleviating Over-segmentation Errors by Detecting Action Boundaries' accepted in WACV2021", "url": "https://github.com/yiskw713/asrf"}, {"title": "Alleviating Over-segmentation Errors by Detecting Action Boundaries", "url": "https://openaccess.thecvf.com/content/WACV2021/papers/Ishikawa\_Alleviating\_Over-Segmentation\_Errors\_by\_Detecting\_Action\_Boundaries\_WACV\_2021\_paper.pdf"}, {"title": "Alleviating Over-segmentation Errors by Detecting Action Boundaries | Request"}]

PDF", "url": "https://www.researchgate.net/publication/342944719\_Alleviating\_Over-segmentation\_Errors\_by\_Detecting\_Action\_Boundaries"}, {"title": "WACV 2021 Open Access Repository", "url": "https://openaccess.thecvf.com/content/WACV2021/html/Ishikawa\_Alleviating\_Over-Segmentation\_Errors\_by\_Detecting\_Action\_Boundaries\_WACV\_2021\_paper.html"}]}

Based on the search results, here's information about the ASRF paper:

The Action Segment Refinement Framework (ASRF) is an effective framework for temporal action segmentation that consists of a long-term feature extractor and two branches: the Action Segmentation Branch (ASB) and the Boundary Regression Branch (BRB).

The long-term feature extractor provides shared features for the two branches with a wide temporal receptive field, with the ASB classifying video frames with action classes, while the BRB regresses the action boundary probabilities. The action boundaries predicted by the BRB refine the output from the ASB, which results in a significant performance improvement.

The paper was authored by Yuchi Ishikawa, Seito Kasai, Yoshimitsu Aoki, and Hirokatsu Kataoka, and was published in the Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) in 2021.

The model outperforms state-of-the-art methods on three challenging datasets, offering an improvement of up to 13.7% in terms of segmental edit distance and up to 16.1% in terms of segmental F1 score.

The official code repository is available on GitHub at <https://github.com/yiskw713/asrf>.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

## 12. user (2026-06-16T03:23:44.969Z)

Web search results for query: "Action Segment Refinement Framework ASRF two branches action classification boundary regression architecture WACV 2021"

Links: [{"title": "Alleviating Over-segmentation Errors by Detecting Action Boundaries", "url": "https://yiskw713.github.io/asrf/"}, {"title": "177 - Alleviating Over-segmentation Errors by Detecting Action Boundaries - Crossminds", "url": "https://crossminds.ai/video/177-alleviating-over-segmentation-errors-by-detecting-action-boundaries-602f247b96aa98bc22b14172/"}, {"title": "Alleviating Over-segmentation Errors by Detecting Action Boundaries", "url": "https://arxiv.org/pdf/2007.06866"}, {"title": "Alleviating Over-segmentation Errors by Detecting Action Boundaries | DeepAI", "url": "https://deepai.org/publication/alleviating-over-segmentation-errors-by-detecting-action-boundaries"}, {"title": "Alleviating Over-Segmentation Errors by Detecting Action Boundaries", "url": "https://papertalk.org/papertalks/27315"}, {"title": "Alleviating Over-segmentation Errors by Detecting Action Boundaries | Request PDF", "url": "https://www.researchgate.net/publication/352386005\_Alleviating\_Over-segmentation\_Errors\_by\_Detecting\_Action\_Boundaries"}, {"title": "GitHub - yiskw713/asrf: Code for 'Alleviating Over-segmentation Errors by Detecting Action Boundaries' accepted in WACV2021", "url": "https://github.com/yiskw713/asrf"}, {"title": "Alleviating Over-segmentation Errors by Detecting Action Boundaries | IEEE Conference Publication | IEEE Xplore", "url": "https://ieeexplore.ieee.org/document/9423184/?jsessionid=Xz0V0pGDMQo2CoSEFgaEYyqsH0b5syMaSpA9dHe6VYnMhZ7kXxLr!1048047913"}, {"title": "Alleviating Over-segmentation Errors by Detecting Action Boundaries", "url": "https://openaccess.thecvf.com/content/WACV2021/papers/Ishikawa\_Alleviating\_Over-Segmentation\_Errors\_by\_Detecting\_Action\_Boundaries\_WACV\_2021\_paper.pdf"}, {"title": "asrf/evaluate.py at main · yiskw713/asrf", "url": "https://github.com/yiskw713/asrf/blob/main/evaluate.py"}]}

Based on the search results, I found comprehensive information about the Action Segment Refinement Framework (ASRF) paper from WACV 2021:

## Overview

The Action Segment Refinement Framework (ASRF) was proposed for the temporal action segmentation

task and presented at WACV 2021.

## Architecture

The model architecture consists of a long-term feature extractor and two branches: the Action Segmentation Branch (ASB) and the Boundary Regression Branch (BRB). The long-term feature extractor provides shared features for the two branches with a wide temporal receptive field.

## Two Branches and Their Functions

The ASB classifies video frames with action classes, while the BRB regresses the action boundary probabilities. The action boundaries predicted by the BRB refine the output from the ASB, which results in a significant performance improvement.

## Key Contributions

The framework divides temporal action segmentation into frame-wise action classification and action boundary regression. The framework outperforms state-of-the-art methods on three challenging datasets, offering an improvement of up to 13.7% in terms of segmental edit distance and up to 16.1% in terms of segmental F1 score.

The official implementation is available on GitHub at <https://github.com/yiskw713/asrf>.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

### 13. assistant (2026-06-16T03:23:48.360Z)

Now I have the key facts. Let me analyze the claim carefully against the primary ASRF architecture to assess the strong assertion (model-independent / usable as a post-hoc splitter).

Let me verify the precise mechanism by reading the ASRF paper's own description of how the boundaries combine with classification, to test whether "post-hoc splitter on an existing dense signal" is a fair reading.

### 14. user (2026-06-16T03:23:54.707Z)

I apologize, but the provided content appears to be a corrupted or binary PDF file that cannot be reliably parsed for the specific technical details you're requesting about ASRF's architecture.

The document displays as raw PDF stream data rather than readable text, making it impossible to:

- Identify how ASB and BRB components interact
- Determine if boundary regression is a post-processing step or jointly-trained
- Extract quotes about the refinement procedure
- Clarify whether the approach is model-independent

To answer your questions about ASRF accurately, I would need:

1. A properly formatted/OCR'd version of the paper, or
2. The paper's title and authors so I could reference the published version, or
3. Plain text content from the original document

Could you provide the paper in a more accessible format?

[Binary content (application/pdf, 1.4MB) also saved to C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\tool-results\webfetch-1781579908809-uip056.pdf]

### 15. assistant (2026-06-16T03:23:56.636Z)

Let me extract the ASRF primary paper text directly to verify the refinement mechanism.

#### 16. user (2026-06-16T03:24:02.294Z)

=== [post-process] ===

This shows that our framework not only re?nes frame-level ac- tion predictions with boundaries, but selects reasonable ac- tion boundaries during re?nement as well. Therefore, the ASB and the BRB are mutually supportive. F1@{10, 20, 50} Edit Acc w/o post-processing 76.1 74.5 66.7 68.5 82.6 Relabeling (?t = 5) 81.5 79.7 71.6 74.8 82.6 Relabeling (?t = 15) 82.4 80.6 72.5 76.2 82.7 Smoothing (K = 5) 80.7 78.9 70.8 73.5 82.6 Smoothing (K = 15) 80.8 79.0 70.9 73.6 82.6 Similarity 39.8 30.9 18.1 32.8

=== [end-to-end] ===

uage of actions: Recovering the syntax and seman- tics of goaldirected human activities. IEEE/CVF Inter- national Conference on Computer Vision and Pattern Recognition (CVPR), pages 780?787, 2014. [23] Hilde Kuehne, Juergen Gall, and Thomas Serre. An end-to-end generative framework for video segmen- tation and recognition. IEEE Winter Conference on Applications of Computer Vision (WACV), pages 1?8, 2016. [24] Hilde Kuehne, Alexander Richard, and Juergen Gall. Weakly supervised learning of actions

=== [jointly] ===

ion sequence with a Markov model [23, 36] or an RNN [39] to classify frame- wise actions. Lea et al. [26] propose a spatiotemporal CNN for extracting features, which takes RGB and motion his- tory images as input, and uses a semi-Markovian model for jointly segmenting and classifying actions. Following the success in the speech synthesis domain, some studies have adopted a temporal convolutional net- work from the WaveNet model [46]. Lea et al. [25] propose two types of temporal convolutional net

#### 17. assistant (2026-06-16T03:24:04.483Z)

Let me get the precise refinement procedure and check for any explicit "model-agnostic / applicable to other models" claim in the ASRF paper itself.

#### 18. user (2026-06-16T03:24:11.657Z)

**refine the -> hits 0**

**refinement -> hits 0**

**agnostic -> hits 1**

**other models -> hits 0**

**existing -> hits 7**

**our framework -> hits 29**

**feature extractor -> hits 12**

=== [agnostic] === ng the action segmentation results from the ASB (see Section 3.4). Action boundaries are de?ned as frames when an ac- tion starts and ends irrespective of action classes. Unlike the methods which use a Hidden Markov model to determine the most probable sequence of actions [23, 42], the BRB is class-agnostic, which eliminates the need for modeling the probabilities from each class to every other. Rather, the BRB only regresses the likelihood of general action bound- aries. Therefore, the BRB requires far less data for train- ing and can improve the robustness in comparison with the class-aware

#### 19. assistant (2026-06-16T03:24:19.272Z)

This is the decisive evidence. The ASRF paper's only use of "agnostic" is \*\*\*"the BRB is class-agnostic"\*\*\* ? meaning it regresses general boundary likelihood irrespective of action class. The paper does NOT describe ASRF as "model-agnostic" or "applicable to any temporal segmentation output." Let me confirm there is no explicit model-independence / plug-onto-other-models claim, and capture the actual refinement procedure (how BRB boundaries split ASB output).

#### 20. user (2026-06-16T03:24:24.515Z)

==== HEAD (first 1600 chars) ====

Alleviating Over-segmentation Errors by Detecting Action Boundaries Yuchi Ishikawa<sup>1,2</sup>, Seito Kasai<sup>1,2</sup>, Yoshimitsu Aoki<sup>2</sup>, Hirokatsu Kataoka<sup>1</sup> <sup>1</sup>National Institute of Advanced Industrial

Science and Technology (AIST) 2Keio University {yuchi.ishikawa, seito-kasai, hirokatsu.kataoka}@aist.go.jp, aoki@elec.keio.ac.jp Abstract We propose an effective framework for the temporal action segmentation task, namely an Action Segment Refinement Framework (ASRF). Our model architecture consists of a long-term feature extractor and two branches: the Action Segmentation Branch (ASB) and the Boundary Regression Branch (BRB). The long-term feature extractor provides shared features for the two branches with a wide temporal receptive field. The ASB classifies video frames with action classes, while the BRB regresses the action boundary probabilities. The action boundaries predicted by the BRB refine the output from the ASB, which results in a significant performance improvement. Our contributions are three-fold: (i) We propose a framework for temporal action segmentation, the ASRF, which divides temporal action segmentation into frame-wise action classification and action boundary regression. Our framework refines frame-level hypotheses of action classes using predicted action boundaries. (ii) We propose a loss function for smoothing the transition of action probabilities, and analyze combinations of various loss functions for temporal action segmentation. (iii) Our framework outperforms state-of-the-art methods on three challenging datasets, offering an improvement of up to 13.7

=== refine combine === shared features for the two branches with a wide temporal receptive field. The ASB classifies video frames with action classes, while the BRB regresses the action boundary probabilities. The action boundaries predicted by the BRB refine the output from the ASB, which results in a significant performance improvement. Our contributions are three-fold: (i) We propose a framework for temporal action segmentation, the ASRF, which divides temporal action segmentation into frame-wise action classification and action boundary regression. Our framework refines frame-level hypotheses of action classes

=== contributions === the BRB refines the output from the ASB, which results in a significant performance improvement. Our contributions are three-fold: (i) We propose a framework for temporal action segmentation, the ASRF, which divides temporal action segmentation into frame-wise action classification and action boundary regression. Our framework refines frame-level hypotheses of action classes using predicted action boundaries. (ii) We propose a loss function for smoothing the transition of action probabilities, and analyze combinations of various loss functions for temporal action segmentation. (iii) Our framework outperforms state-of-the-art methods on three challenging datasets, offering an improvement of

## 21. assistant (2026-06-16T03:24:42.522Z)

I now have complete evidence. Let me assess the claim against everything gathered.

### Analysis

**The claim has three parts:**

1. ASRF reduces over-segmentation by adding a complementary network branch that directly regresses action-boundary locations ? **TRUE** (confirmed by the primary ASRF paper: BRB regresses boundary probabilities; title is literally "Alleviating Over-segmentation Errors by Detecting Action Boundaries").
2. The boundary-refinement approach is "model-independent" ? **stated by the cited survey**, but this is the survey's characterization, not the ASRF paper's own claim.
3. "It can be applied to the output of any temporal segmentation model (i.e. usable as a post-hoc splitter on an existing dense signal)" ? this is the claim's **interpretive gloss**, and it's where the overreach lies.

**Key findings on the checklist:**

1. **Supported by quote?** The supporting quote is **verbatim present** in the cited survey (2210.10352, Ding/Sener/Yao). So the quote?claim mapping is faithful. Good.
2. **Is the survey's characterization itself accurate?** Here is the problem. The primary ASRF paper (Ishikawa et al., WACV 2021, arXiv 2007.06866) describes ASRF as a **jointly-trained dual-branch network**: a **shared** long-term feature extractor feeds an Action Segmentation Branch (ASB) and a Boundary Regression Branch (BRB). The boundaries refine the ASB's **own** output. The paper's only use of "agnostic" is **"the BRB is class-agnostic"** (boundaries irrespective of action class) ? NOT model-agnostic. The ASRF paper makes **no claim** that the branch can be bolted onto any other model's output, nor that it is a post-hoc/post-processing module. ASRF's actual ablations treat boundary use as part of an integrated framework with shared features.



3. **\*\*The "post-hoc splitter on an existing dense signal" gloss is the overreach.\*\*** ASRF is not a post-processing splitter you drop onto a foreign dense signal; it is an end-to-end architecture whose BRB is trained on shared features alongside the classifier it refines. The survey's "model-independent / applicable to any temporal segmentation output" is a loose secondary-source paraphrase (arguably defensible only in the weak sense that "boundary regression" *\*as a general idea\** is reusable), but the claim escalates it into a concrete engineering capability ("usable as a post-hoc splitter on an existing dense signal") that the primary source does not support.

**\*\*Verdict on source-quality vs claim-strength:\*\*** The strong, operational part of the claim (drop ASRF onto any existing dense signal as a post-hoc splitter) rests on a secondary-source paraphrase that the primary source contradicts in mechanism. For an engineering-actionable strong claim, the primary source is authoritative and it describes a jointly-trained, shared-feature architecture ? not a model-independent post-processor.

This refutes the load-bearing part of the claim (the post-hoc/model-independent applicability). The first part (boundary-regression branch reduces over-segmentation) is solidly true, but the claim as written packages that with a misleading capability assertion.

This meets the refutation bar: the strong assertion is contradicted by the primary source's actual architecture.

## **22. user (2026-06-16T03:24:52.974Z)**

Structured output provided successfully

## **23. assistant (2026-06-16T03:24:58.691Z)**

Verification complete. Verdict: **\*\*refuted = true\*\*** (high confidence).

Summary of findings:

- The supporting quote is genuinely present verbatim in the cited survey (`arXiv:2210.10352`, Ding/Sener/Yao), so the transcription is faithful and the source is high-quality.
- Part 1 of the claim (ASRF adds a boundary-regression branch to reduce over-segmentation) is true per the primary paper.
- The load-bearing strong assertion ? "model-independent," "applicable to any temporal segmentation model," "usable as a post-hoc splitter on an existing dense signal" ? is an overreach contradicted by the primary ASRF paper (`arXiv:2007.06866`, Ishikawa et al., WACV 2021). ASRF is a jointly-trained dual-branch network sharing a long-term feature extractor; the BRB refines the ASB's *\*own\** output. The paper's only "agnostic" claim is that the BRB is **\*\*class\*\***-agnostic, not model-agnostic. The "model-independent / post-hoc splitter" framing is the survey's loose secondary paraphrase, not a capability the primary source supports.